DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

Linear Algebra

Eigen Values and CHT

Lecture No. 02



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Recap of previous lecture



Topic

EIGEN VALUES & their Properties

(BASICS)

Topics to be Covered



Topic

Remaining Properties of Eigen Values

RECAP:

① For Sq. Mat $A_{n\times n}$ if $\boxed{AX=\lambda X}$ → E. Vector, → E. Value

② E. Vector ← Non Zero.

③ λ ← Real or Complex.

④ If $\lambda=0 \Rightarrow AX=\lambda X=0\cdot X=O$

i.e. $AX=O$ i.e. X Converts into Point Vector

⑤ C. Equⁿ of A is $|A-\lambda I|=0$

⑥ Let $X=0$ then $AX=A\cdot 0=0=\lambda\cdot 0=\lambda X$

Nothing special about $X=0$ so E. Vector can not be a ZERO VECTOR, But E Value may be zero.

PROPERTIES of E. Values → Let $A_{n\times n}$ is the given Matrix

& let $\lambda_1, \lambda_2, \lambda_3, \ldots\ldots, \lambda_n$ are it's E. Values.

① Number of E. Values of A = order of A (whether distinct or Repeated)

② Sum of E. Values = Trace (A) i.e $\lambda_1+\lambda_2+\lambda_3+\ldots\ldots+\lambda_n = Tr(A)$

③ Product of E Values = Det (A) i.e $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 \ldots\ldots \cdot \lambda_n = |A|$

④ (Zero is an E. Value of A) iff (Mat is Singular) i.e $\lambda = 0 \iff |A| = 0$

⑤ E Values of U.T.M, L.T.M, Diagonal Mat, Scalar Mat are just the Diag elements

eg $A = \begin{bmatrix} 2 & -1 & 0 & 4 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 4 \end{bmatrix} \Rightarrow \lambda = 2, 3, 0, 4$

② $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} \Rightarrow \lambda = 1, -1, 2, 4$

(6) W.K.T that EValues of $A_{n\times n}$ are $\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n$ then

(i) EValues of A^T are same i.e $\lambda_1, \lambda_2, \lambda_3 \ldots, \lambda_n$

(ii) EValues of A^m are $\lambda_1^m, \lambda_2^m, \lambda_3^m, \ldots, \lambda_n^m$ (where $m \in N$)

(iii) EValues of A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}, \ldots, \frac{1}{\lambda_n}$ (Provided $|A| \neq 0$)

(iv) EValues of (adj A) are $\frac{|A|}{\lambda_1}, \frac{|A|}{\lambda_2}, \frac{|A|}{\lambda_3}, \ldots, \frac{|A|}{\lambda_n}$

Proof (ii)

$AX = \lambda X$ → EValue of A

$A(AX) = \lambda(AX)$

$A^2X = \lambda(\lambda X)$

$A^2X = \lambda^2 X$ → EValue of A^2

Proof (iii)

$AX = \lambda X$ → EValue of A

$\lambda X = AX$

$\lambda(A^{-1}X) = A^{-1}(AX)$

$A^{-1}X = \left(\frac{1}{\lambda}\right)X$ → EValue of A^{-1}

(7) Don't use E-operations in a given Mat while calculating E. Values. But we can apply 3rd E-operation in id's C-Equ's.

ie Equivalent Matrices may have different E. Values, Although they have same Rank

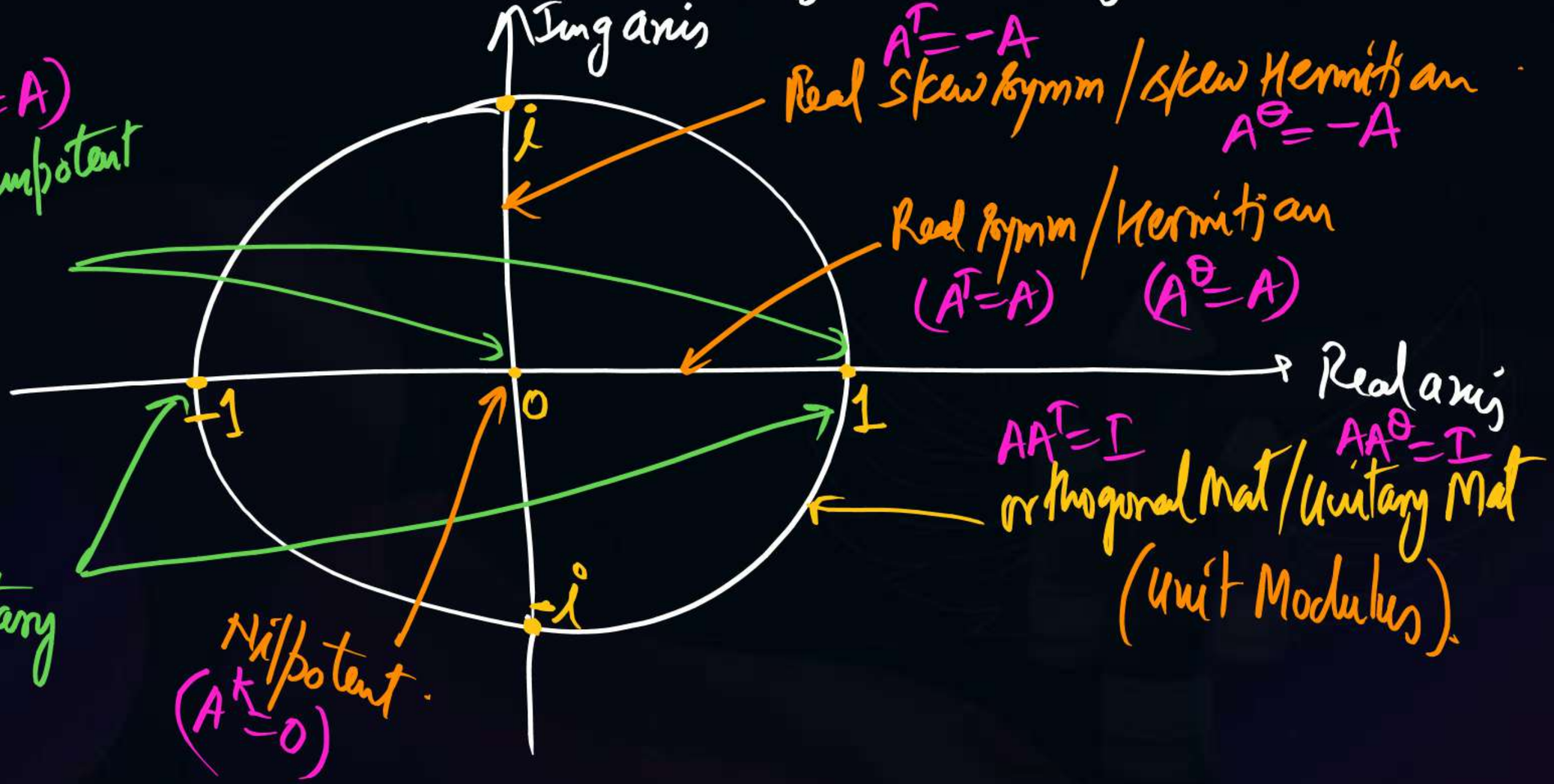
(8) If in a Mat, Sum of all the elements in each Row (or in each Column) is unique Const K then that Constant (K) will be one of the E Value of A

(9) Number of Non Zero E Values of $A \leq \rho(A)$

(10) If λ is an E value of A then to find E Value of any algebraic expression formed by A, we can Replace A with λ in that expression.

Shortcut Method to Learn Various Theorems → $|i|=1, |-i|=1$

Consider a unit circle centered at origin i.e $x^2+y^2=1$



eg: E Values of Real Symm Mat are Real & +ve (F)

T F

eg $A=\begin{bmatrix} 2 & -5 & 4 \\ -5 & 3 & 1 \\ 4 & 1 & 0 \end{bmatrix}$, $A=\begin{bmatrix} 2 & -5 & 4i \\ -5 & 0 & 2+3i \\ 4i & 2+3i & 7 \end{bmatrix}$

= Real Symm. = Comp. Symm.

⇓ ⇓

$\lambda \in R$ Not Necessarily Real

Real E values ⇒ Either A is Real Symm or A is Hermitian (∵ a_{12} = Comp)

Qe If $A=\begin{bmatrix} 2 & 5+i & -3 \\ x & 4 & 1 \\ -3 & 1 & 2 \end{bmatrix}$ has only Real E. Values then $x=?$

Hence $x=5-i$.

eg Idempotent Mat: $A^2 = A$

$\Rightarrow \lambda^2 = \lambda$

$\lambda(\lambda-1) = 0$

$\lambda = 0$ or 1

eg Involutary Mat: $A^2 = I$

$\Rightarrow \lambda^2 = 1$

$\lambda = \pm 1$

is $\lambda = 1$ or -1

Let $A_{3\times3}$ s.t A is Involutary Mat then Possible set of EValues of A will be?

MSQ

(a) $\{1,1,1\}$

(b) $\{-1,-1,-1\}$

(c) $\{1,1,-1\}$

(d) $\{1,-1,-1\}$

(e) $\{1,0,-1\}$

Note: for $A_{2\times2}$ s.t $A^2 = I$ then possible set of EValues of A will be?

(a) $\{1,1\}$ (b) $\{-1,-1\}$

(c) $\{1,-1\}$, (d) $\{2,0\}$

Q. if $A_{n\times n}$ s.t $\{\lambda_i, X_i\}$ is an E. pair of $A_{n\times n} X_{n\times 1} = \lambda X_{n\times 1}$ then which is false

(a) If Non Zero E. vector if $\rho(A-\lambda I) < n$ (T)

(b) if $A^{\theta} = A$ then $\lambda_i \in R \ \forall i$ T ($\because$ A is Hermitian)

(c) if $\bar{A}^{-1} = A^T$ then $|\lambda_i| = 1 \ \forall i$ (T) ($\because$ A is orthogonal Mat)

(d) $\{\lambda_i^m, X_i^m\}$ is an E. Pair of A^m (F) $\because X^2 = XX = \begin{bmatrix} & \\ & \end{bmatrix}_{n\times 1}\begin{bmatrix} & \\ & \end{bmatrix}_{n\times 1} = \text{ND}$

T F

(e) if $A = \bar{A}^{-1}$ then the E. Value of A will be 1 (T) ($\because$ A is an Involutary Mat)

Doubt: $AX = \lambda X$

$(A - \lambda I)X = 0$

$MX = 0$ — (1)

↓

Non Zero EVector

Non Zero Sol.

∞ Sol.

$\rho(M) < n$ ✓ $\quad |M| = 0$

$\rho(A - \lambda I) < n$ or $|A - \lambda I| = 0$

Q: if $AA^{\theta} = I$ ⇒ A is Unitary Mat. then which of the following can not be the E Value of A

(a) $-1 \Rightarrow |\lambda| = |-1| = 1$ ✓

(b) $\frac{-1}{2} + i\frac{\sqrt{3}}{2} \Rightarrow |\lambda| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$ ✓

(c) $\frac{\sqrt{3}}{2} - \frac{i}{2} \Rightarrow |\lambda| = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$ ✓

(d) $\frac{1}{4} + i\frac{\sqrt{3}}{4} \Rightarrow |\lambda| = \sqrt{\frac{1}{16} + \frac{3}{16}} = \frac{1}{2}$ ✗

$z = x + iy \Rightarrow |z| = \sqrt{x^2 + y^2}$

Q: if for any Mat $A_{2\times2}$, $a_{11}=a_{12}=a_{21}=1$ & $a_{22}=-1$ then EValues of A^{19} = ?

(a) $\pm\sqrt{2}$

(b) ± 512

(c) $\pm 512\sqrt{2}$

(d) None

$$A=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}=\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

C.Equ^{n} of A is $|A-\lambda I|=0$

.....

$$\lambda^2-2=0$$

$$\lambda=\pm\sqrt{2}$$

i.e EValues of A are $\sqrt{2}$ & $-\sqrt{2}$

$\therefore$ $\because$ EValues of A^{19} are $(\sqrt{2})^{19}$ & $(-\sqrt{2})^{19}$

i.e $(\sqrt{2})^{18}\sqrt{2}$ & $(-\sqrt{2})^{18}(-\sqrt{2})$

$2^9\sqrt{2}$ & $-2^9\sqrt{2}$

$512\sqrt{2}$ & $-512\sqrt{2}$

i.e $\pm 512\sqrt{2}$ Ans

Q. If λ is one of the E. Value of $A=\begin{bmatrix} 40 & -24 & -16 \\ -11 & 30 & -19 \\ 16 & 34 & -50 \end{bmatrix}$

then another E. Value will be?

(a) $\lambda - 20$

(b) 20λ

(c) ✓ $20 - \lambda$

(d) $20 + \lambda$

$\lambda_1 + \lambda_2 + \lambda_3 = Tr(A)$

$\lambda + \lambda_2 + 0 = 40 + 30 - 50$

$\lambda + \lambda_2 = 20$

$\lambda_2 = (20 - \lambda)$

$\because |A| = \begin{vmatrix} 40 & -24 & -16 \\ -11 & 30 & -19 \\ 16 & 34 & -50 \end{vmatrix}$

$C_1 \to C_1 + C_2 + C_3 \Rightarrow \begin{vmatrix} 0 & -24 & -16 \\ 0 & 30 & -19 \\ 0 & 34 & -50 \end{vmatrix} = 0$

$\because |A| = 0 \Rightarrow$ one of the E.value $= 0$.

Q One of the E. Value of

$$A=\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{bmatrix}$$ will be ?

(10) (10) (10) (10)

(M-I) By Prop (8) ; $\lambda = 10$ Ans

(M-II) C.Eq of A is $|A-\lambda I|=0$

$$\begin{vmatrix} (1-\lambda) & 2 & 3 & 4 \\ 2 & (3-\lambda) & 4 & 1 \\ 3 & 4 & (1-\lambda) & 2 \\ 4 & 1 & 2 & (3-\lambda) \end{vmatrix} = 0$$

$C_1 \to C_1+(C_2+C_3+C_4)$

$$\begin{vmatrix} 10-\lambda & 2 & 3 & 4 \\ 10-\lambda & 3-\lambda & 4 & 1 \\ 10-\lambda & 4 & 1-\lambda & 2 \\ 10-\lambda & 1 & 2 & 3-\lambda \end{vmatrix} = 0$$

$$(10-\lambda)\begin{vmatrix} 1 & 2 & 3 & 4 \\ 1 & 3-\lambda & 4 & 1 \\ 1 & 4 & 1-\lambda & 2 \\ 1 & 1 & 2 & 3-\lambda \end{vmatrix} = 0$$

$(10-\lambda)$ [Cubic equation in λ] $=0$

$\Rightarrow \lambda = 10$ & $\lambda = \lambda_2, \lambda_3, \lambda_4$

Ans

Note If $A=\begin{bmatrix}1&2&3&4&5\\2&3&4&5&1\\3&4&5&1&2\\4&5&1&2&3\\5&1&2&3&4\end{bmatrix}$ has only one Real E-Value then find it ?

$= (15)(15)(15)(15)(15)$

(a) 10 (b) 15

(c) 0 (d) None

Sol By Prop (8), $\lambda = 15$ (Real)

Analysis :- $\lambda_1+\lambda_2+\lambda_3+\lambda_4+\lambda_5 = Tr(A)$

$\lambda + (\alpha+i\beta)+(\alpha-i\beta)+(\gamma+i\delta)+(\gamma-i\delta) = 15$

$\lambda + 2\alpha + 2\gamma = 15$

$\lambda + 2(\alpha+\gamma) = 15$

when $\alpha = -\gamma$ then $\lambda = 15$

i.e Nature of E Values are

$\lambda = 15,\ \alpha \pm i\beta,\ -\alpha \pm i\delta$ Ans

Qs The EValues of $A=\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}$?

(WRONG APP)

M-I $R_2 \to R_2-R_1,\ R_3 \to R_3-R_1,\ R_4 \to R_4-R_1$

$A \sim \begin{bmatrix}1&1&1&1\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix}$ = U.T.M.

$\therefore \lambda = 1, 0, 0, 0$

$\because$ it is a Voilation of Prop (7).

M-II $A=\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}$

$= 4\ 4\ 4\ 4$

By Prop (8), one EValue is $\lambda = 4$.

Now $\rho(A) =$ one

$\therefore$ No. of Non Zero EValues $\le \rho(A)$ — By Prop (9)

i.e ≤ 1

it means A has only one Non Zero EValue (which is $\lambda=4$) & the Remaining EValues are $\lambda = 0, 0, 0$

Hence EValues $= 4, 0, 0, 0$

M-II Using Conventional App →

C. Equ^n is $|A-\lambda I|=0$

$$\begin{vmatrix} (1-\lambda) & 1 & 1 & 1 \\ 1 & (1-\lambda) & 1 & 1 \\ 1 & 1 & (1-\lambda) & 1 \\ 1 & 1 & 1 & (1-\lambda) \end{vmatrix} = 0$$

$$\begin{vmatrix} (4-\lambda) & 1 & 1 & 1 \\ (4-\lambda) & 1-\lambda & 1 & 1 \\ (4-\lambda) & 1 & 1-\lambda & 1 \\ (4-\lambda) & 1 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1-\lambda & 1 & 1 \\ 1 & 1 & 1-\lambda & 1 \\ 1 & 1 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)\begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -\lambda & 0 & 0 \\ 0 & 0 & -\lambda & 0 \\ 0 & 0 & 0 & -\lambda \end{vmatrix} = 0$$

$(4-\lambda)(-\lambda^3)=0 \Rightarrow \lambda = 0, 0, 0, 4$

M-IV $A=$ 

By Prop (8), $\lambda=4$ & Now $\lambda_1+\lambda_2+\lambda_3+\lambda_4=Tr$

Now there is possibility that $\lambda_2=\alpha, \lambda_3=-\alpha$ & $\lambda_4=0$ i.e. $4+\alpha-\alpha+0=4$ i.e. $0=0$

that is why it is not a very good method.

Q. if $A=\begin{bmatrix}4&2\\2&4\end{bmatrix}$ then find E.Values of $(A^2-5A+2I)=?$

Sol: Let $B=A^2-5A+2I$

$$=\begin{bmatrix}4&2\\2&4\end{bmatrix}\begin{bmatrix}4&2\\2&4\end{bmatrix}-5\begin{bmatrix}4&2\\2&4\end{bmatrix}+2\begin{bmatrix}1&0\\0&1\end{bmatrix}$$

$$=\begin{bmatrix}20&16\\16&20\end{bmatrix}-\begin{bmatrix}20&10\\10&20\end{bmatrix}+\begin{bmatrix}2&0\\0&2\end{bmatrix}$$

$$=\begin{bmatrix}2&6\\6&2\end{bmatrix}$$

C. Equ^n of B is $|B-\lambda I|=0$

$$\lambda^2-4\lambda-32=0$$

$$(\lambda-8)(\lambda+4)=0 \Rightarrow \lambda=8,-4$$

M-II By Prop (10),

$A=\begin{bmatrix}4&2\\2&4\end{bmatrix}$ $\lambda=6$, $\lambda=2$

YESTERDAY it was 1st Question

$B=(A^2-5A+2I)$ $6^2-5(6)+2=8$, $2^2-5(2)+2=-4$

$\therefore$ E.Values of B are 8 & -4

Q. (TAGDAA Question) If EValues of $X_{2\times2}$ are -2 & -3 then EValues of $B=(X+I)^{-1}(X+5I)$?

M-I

$X \begin{cases} -2 \\ -3 \end{cases}$ then $B=(X+I)^{-1}(X+5I)$

$=(-2+1)^{-1}(-2+5)=\frac{3}{(-1)}=-3$

$=(-3+1)^{-1}(-3+5)=\frac{2}{(-2)}=-1$

M-II Let $X=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where

$a+d=(-2)+(-3)=-5$

$ad-bc=(-2)(-3)=6$

$T_r=a+d$ & $|X|=ad-bc$

so it is not possible to find unique values of a, b, c, d & Hence X

it is an underdetermined Solution

I.e Data Inadequate

Q-III Let $X=\begin{bmatrix}-2 & 0\\ 0 & -3\end{bmatrix}$

$X+I=\begin{bmatrix}-1 & 0\\ 0 & -2\end{bmatrix}$, $(X+I)^{-1}=\frac{1}{2}\begin{bmatrix}-2 & 0\\ 0 & -1\end{bmatrix}$

$X+5I=\begin{bmatrix}3 & 0\\ 0 & 2\end{bmatrix}$

Now $B=(X+I)^{-1}(X+5I)$

$=\frac{1}{2}\begin{bmatrix}-2 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 2\end{bmatrix}=\ldots\ldots=\begin{bmatrix}-3 & 0\\ 0 & -1\end{bmatrix}$

So E Values of B are -3 & -1.



THANK - YOU

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